

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • About 8 nights (1) • The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly (1)	2
01.2	<i>1 mark for each point.</i> • Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size (1) • Sampling at different conditions / times would make results not comparable (1) • Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season) (1)	3
01.3	<i>1 mark for each point.</i> • Accuracy: how close a recorded result is to the real value (1) • Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm) (1)	2
01.4	<i>1 mark for each point.</i> • A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level (1) • Standardised techniques ensure results collected at different times, places or by different people are comparable (1) • Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned (1) • HCFCs to be phased out (by 2030); some essential uses (eg halon fire extinguishers in aircraft) still permitted (1) • A fund to help (developing) countries implement the Protocol; legally binding targets (unlike the 1985 Vienna Convention) (1)	3
02.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Aerosols: pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk (2) • Refrigerants: HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases (2)	4
02.3	<i>1 mark for each point.</i> • CFCs are drained from the appliance and incinerated / broken down at high temperature (1) • Products are CO₂ and acidic gases (HCl, HF) which are neutralised with crushed lime (1)	2
02.4	<i>1 mark.</i> • CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years; some ODSs (eg dichloromethane) are not banned (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	
03.1	<i>1 mark for each point.</i> • High mean wind velocities that are consistent in strength and direction (1) • A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK (1)	2
03.2	<i>1 mark for each point.</i> • Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber; the air forced in and out drives a turbine (1) • Or point absorber (floating buoy moving relative to a fixed base turns a generator); surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine); overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines); oscillating wave surge converter (Oyster) (1)	2
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Equipment must withstand storms and corrosion (1) • Difficult to anchor off stormy coasts / in deep water (1) • Expensive to transmit electricity from isolated coasts to consumers (1) • Competition with established technologies (1)	2

Question	Marking guidance	Marks
03.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Barrage: huge quantities of construction materials; reduces tidal range so intertidal mudflats used by wading birds are lost; changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity; pollutants concentrate in the lagoon; blocks migratory fish and shipping (up to 2) • Lagoon: similar but smaller scale; only part of the estuary enclosed so the most sensitive areas and main channel can be avoided (up to 2) • In-stream turbines: do not block the tidal flow so little impact on range or sediment; but noise affects cetaceans and many dispersed turbines could have a larger total impact; harness less energy (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
04.1	<i>Correct answer only.</i> • B - tiny pores between clay particles allow capillary rise; none in sand (1)	1
04.2	<i>1 mark for each point.</i> • Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants (1) • Capillary: held in pores by surface tension; does not drain but can be absorbed by plants (1) • Gravitational: drains freely through large pores; available to plants only briefly as it passes (1)	3
04.3	<i>1 mark.</i> • Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
05.1	<i>1 mark for each point.</i> • There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) (1) • Most phosphorus compounds have low solubility so little is available in soil water (1) • Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks (1)	3
05.2	<i>1 mark for each point.</i> • Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil (1) • Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates (1)	2
05.3	<i>1 mark for each point.</i> • Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) (1) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment (1) • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland (1)	3

Question	Marking guidance	Marks
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock (1) • Breed crops that absorb phosphate more efficiently (1) • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) (1) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
06.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(68\,000 - 7\,000) / 68\,000 \times 100$ (1) • = 89.7% (accept 90%) (1)	2
06.2	<i>1 mark for each point.</i> • Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued (1) • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume (1)	2
06.3	<i>1 mark for each point.</i> • Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake (1) • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) (1) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) (1) • Loss of livelihoods; conflict between the countries sharing the rivers (1)	4
06.4	<i>1 mark for each point.</i> • Upstream abstraction for irrigation reduces the flow available downstream (1) • Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
07.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Discharge of tank-washing water from tankers (1) • Oil rig / drilling accidents / leakage during drilling (1) • Pipeline leaks (1) • Leakage from storage tanks / refinery spills (1) • Waste lubricating oil from vehicles / industry washed into drains (1) • Other ship accidents (fuel oil) (1)	3
07.2	<i>1 mark for each point.</i> • Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia (1) • Feathers lose buoyancy / waterproofing so heavily oiled birds drown (1) • Birds swallow toxic oil when preening; time spent cleaning feathers reduces time for feeding themselves / their young (1)	3

Question	Marking guidance	Marks
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen (1) • Toxic components (benzene, xylene) dissolve and poison plankton near the surface (1) • Oil blocks light so algae / phytoplankton cannot photosynthesise (1) • Oil masks odours / narcotises so animals cannot find mates or food by smell (1) • Shore organisms (molluscs, crustaceans) are covered and asphyxiated (1)	2
07.4	<i>1 mark for each point.</i> • Dispersal / dilution by currents and waves is prevented so oil stays concentrated; in open water most spills disperse in days or weeks (1) • Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
08.1	<i>1 mark for the correct answer.</i> • $15 - (-18) = 33\text{ }^{\circ}\text{C}$ (1)	1
08.2	<i>1 mark for each point.</i> • Visible light passes easily through the atmosphere and is absorbed by the surface, which warms (1) • The warm surface emits (long-wave / far) infrared radiation (1) • Greenhouse gases (water vapour, CO_2, methane) absorb infrared better than the average for the atmosphere and convert it to heat (1) • The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction (1)	4
08.3	<i>1 mark.</i> • Any gas that is better at absorbing infrared radiation than the average for the atmosphere (1)	1
08.4	<i>1 mark for each point.</i> • Arriving: mainly ultraviolet, visible light and near infrared (short wavelength); leaving: mainly long-wavelength (far) infrared (1) • Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained (1)	2
08.5	<i>1 mark for each point.</i> • The energy and the processes it drives control climate, ocean currents and the hydrological cycle (1) • These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
09.1	<i>1 mark for each point.</i> • Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) (1) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people (1)	2

Question	Marking guidance	Marks
09.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Physical state of the effluent (solution / solid) (1) • Ocean currents / river flow (1) • Effect of pH and oxygen on solubility (1) • Bioaccumulation and food-chain concentration (1) • Food sources and consumption by local people (eg shellfish, seaweed) (1) • Half-lives of the isotopes (1) • Meteorological conditions for any gaseous release (1) • Nature of the environment - geology, vegetation (1)	4
09.3	<i>1 mark for each point.</i> • Monitoring focuses on the sites most at risk of contamination (1) • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions (1)	2
09.4	<i>1 mark for each point.</i> • It assesses the risk to the people most exposed because of their lifestyle before any health effects occur (1) • If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9
	Model answer - Level 3 (9 marks)	

Question	Marking guidance	Marks
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction <i>and</i> a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
10.2	1 mark. • The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	

Question	Marking guidance	Marks
11.1	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25